Geometric Sequences and Series

Geometric sequences

1 Consider the sequence $3, 6, 12, 24, \ldots$

a State the common ratio.

b Write a formula for the nth term.

c Find a_{10}.

2 In a geometric sequence, $a_2 = 6$ and $a_5 = 48$. Find the common ratio and the first term.

Geometric series

3 Find the sum of the first 8 terms of $5 + 10 + 20 + \cdots$

4 Evaluate each infinite sum, or state that it diverges:

a $18 + 6 + 2 + \frac{2}{3} + \cdots$

b $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \cdots$

c $2 + 3 + 4.5 + \cdots$

5 Write the repeating decimal $0.\overline{7}$ as a fraction using an infinite geometric series.

6 A ball is dropped from 8 m and rebounds to $\frac{3}{4}$ of its previous height on each bounce. Find the total vertical distance it travels (falling and rising) before coming to rest.

Geometric means and mixed sequences

7 Insert two positive geometric means between 4 and 108 (i.e., find b_1, b_2 so that $4, b_1, b_2, 108$ is geometric).

8 Determine whether each sequence is arithmetic, geometric, or neither:

a $2, 6, 18, 54, \ldots$

b $5, 8, 11, 14, \ldots$

c $1, 4, 9, 16, \ldots$

Applications

9 A car worth 80,000 riyals depreciates by 15% per year.

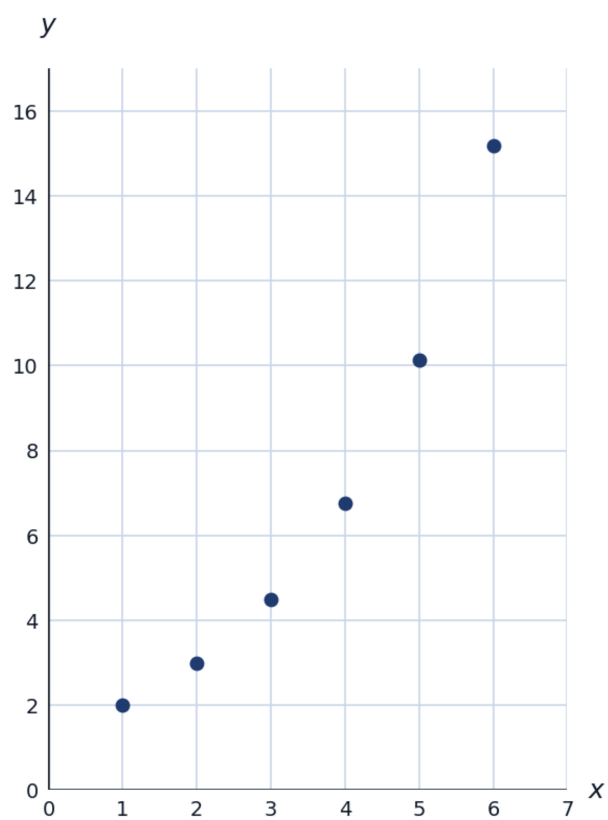
a Write a formula for its value after n years.

b Find its value after 5 years, to the nearest riyal.

10 A chain letter starts with 1 person sending it to 5 people, each of whom sends it to 5 more, and so on. Find the total number of people who have received the letter after 6 rounds (including the original sender's round).

Visualizing geometric growth

11 The first 6 terms of $a_n = 2(1.5)^{n-1}$ are plotted below. Contrast the shape of this graph with the straight-line shape of an arithmetic sequence's graph.



Determining convergence before summing

12 For each infinite geometric series, determine whether it converges WITHOUT computing the sum, by checking $|r| < 1$: $\sum_{n=1}^{\infty} 5(0.9)^{n-1}$, $\sum_{n=1}^{\infty} 3(1.2)^{n-1}$, $\sum_{n=1}^{\infty} 7(-0.4)^{n-1}$.

Finding the number of terms from a partial sum

13 For the geometric series $2+6+18+\cdots$, find the number of terms n needed for the sum to first exceed 2000.

A mixed application: population with geometric decline

14 A population of an endangered species declines by 12% each year from an initial 5000. Find the total number of 'animal-years' lived by the population over the first 10 years (i.e. the sum of the population sizes at the start of each of the 10 years), to the nearest whole number.